TechWise

Smart E-Commerce & Product Intelligence Platform

Project Proposal & Planning Documentation

Team:	Falcon Team
Program:	Digital Egypt Pioneers Initiative (DEPI) — Round 5
Version:	1.0 — Initial Consolidated Draft
Status:	Project Proposal — Draft for Team Review and Approval
Repository:	https://github.com/DEPI-5-Falcon-Project/TechWise_E-Commerce
Purpose:	Project definition, scope, requirements, design direction, implementation planning, risk management, and evaluation criteria.

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Document Control

Field	Details
Project Name	TechWise
Official Description	Smart E-Commerce & Product Intelligence Platform
Team Name	Falcon Team
Program	DEPI — Round 5
Document Type	Project Proposal and Planning Documentation
Repository	https://github.com/DEPI-5-Falcon-Project/TechWise_E-Commerce
Version	1.0 — Initial Consolidated Draft
Status	Draft for Review and Approval
Prepared By	Falcon Team
Project Supervisor	Mohamed Elsokary
Team Member Names	Elshafei Mohamed, Hajar Atef, Shahd Mohammed, Samer Walid, Mustafa Muhammad, Mohamed Alaa El-Din
Final Technology Stack	React + TypeScript | ASP.NET Core Web API + C# | Microsoft SQL Server | AI Provider API + Structured Outputs | Git + GitHub | Testing Tools | Cloud Hosting & Deployment

1. Executive Summary

TechWise is a Smart E-Commerce & Product Intelligence Platform designed to improve how users discover, understand, compare, purchase, manage, upgrade, and resell technology products.

Traditional e-commerce platforms primarily focus on displaying products, managing shopping carts, processing orders, and facilitating transactions. However, purchasing technology products often requires more than browsing product listings. Customers must understand technical specifications, evaluate competing products, verify product information, assess compatibility with existing devices, interpret reviews, and decide whether purchasing a new product or upgrading an existing one represents the better decision.

TechWise aims to address these challenges by combining e-commerce capabilities with product intelligence, AI-assisted shopping, compatibility analysis, structured product information, purchase decision support, and a verified second-life marketplace.

The platform is organized around a complete product lifecycle:

Discover → Understand → Compare → Decide → Buy → Own → Upgrade → Resell → Transfer

The project's core differentiator is its focus on informed purchasing decisions and product lifecycle continuity rather than treating every transaction as an isolated purchase.

TechWise will combine AI-assisted interpretation and explanation with deterministic backend logic. AI will help users express their requirements, understand technical information, explore alternatives, and interpret reviews. The backend and database will remain responsible for factual product records, validation, eligibility checks, compatibility rules, ownership, authorization, and business decisions.

The platform also proposes a second-life marketplace in which eligible products previously purchased through TechWise may be resold while retaining their original product history and purchase-age continuity.

The project will progress through requirements analysis, UI/UX design, architecture and data modeling, implementation, integration, testing, documentation, and final evaluation.

2. Project Background

Technology products are increasingly complex. Smartphones, computers, accessories, networking equipment, and other electronic devices differ in specifications, performance, supported technologies, upgrade paths, and compatibility requirements.

Users may encounter several difficulties when purchasing these products:

- Understanding technical specifications and their practical implications.
- Comparing similar products across multiple dimensions.
- Determining whether a product meets their individual requirements.
- Identifying compatibility issues with products they already own.
- Distinguishing reliable product information from incomplete or misleading claims.
- Interpreting large volumes of customer reviews.
- Evaluating whether to buy a new product, upgrade an existing one, or resell the current device.
- Assessing the history and condition of previously owned products.

TechWise is proposed as an integrated platform that addresses these difficulties through structured product intelligence, decision support, AI-assisted interactions, and lifecycle-aware commerce.

3. Problem Statement

Conventional e-commerce experiences often separate product discovery, technical understanding, comparison, purchasing, ownership management, and resale into different services or disconnected workflows.

As a result, users may need to consult multiple websites, manually interpret specifications, search for compatibility information, and independently evaluate reviews and upgrade options.

Furthermore, second-hand marketplaces may not provide a consistent relationship between a physical device, its original purchase history, its successive owners, and its resale eligibility.

These limitations can make purchasing decisions more difficult and reduce transparency throughout the product lifecycle.

TechWise proposes an integrated solution that connects product information, decision support, purchasing, ownership records, and eligible second-life transactions within one platform.

4. Proposed Solution

TechWise will provide a unified e-commerce and product intelligence experience through the following capabilities:

1. AI Shopping Intelligence for interpreting user needs and assisting product discovery.
2. Verified Second-Life Marketplace for eligible products with platform purchase-origin and ownership conditions.

3. Product Compatibility Graph for representing and evaluating relationships between products and components.
4. Purchase Decision Engine for comparing products against structured requirements and decision criteria.
5. Product Truth Layer for presenting structured product facts with clear evidence and confidence distinctions.
6. Buy vs Upgrade vs Resell Engine for supporting lifecycle-aware purchasing and ownership decisions.

Advanced Smart Filtering will operate as a cross-cutting capability across product discovery, comparison, and decision workflows rather than as a separate core feature.

The solution will follow a layered engineering approach in which user interfaces communicate with backend APIs, business logic, services, and data-access components. AI-assisted functionality will integrate into this architecture without replacing deterministic validation or database-backed facts.

5. Vision, Mission, and Objectives

5.1 Vision

To create a smarter and more transparent e-commerce experience in which users can make informed technology purchasing decisions and maintain continuity throughout the lifecycle of their products.

5.2 Mission

To combine e-commerce, product intelligence, AI-assisted guidance, compatibility evaluation, structured product information, and lifecycle-aware resale into a unified platform.

5.3 Project Objectives

The project aims to:

- Simplify the discovery of products that match user needs.
- Make technical specifications easier to understand.
- Support structured, multidimensional product comparisons.
- Improve purchasing decisions using explicit requirements and product facts.
- Represent product compatibility through structured relationships.
- Improve transparency regarding product information and evidence.
- Connect purchasing with ownership and lifecycle management.
- Establish eligibility rules for a platform-origin second-life marketplace.
- Support comparisons between purchasing, upgrading, and reselling.
- Deliver a consistent, responsive interface for desktop and mobile users.
- Demonstrate responsible integration between AI-assisted features and deterministic backend systems.

6. Target Users and Stakeholders

6.1 Primary Users

A. Technology Buyers

Users seeking to discover, evaluate, compare, and purchase technology products.

B. Users With Specific Requirements

Users who need assistance identifying products that satisfy defined budgets, technical requirements, preferences, or compatibility constraints.

C. Existing Product Owners

Users who want to understand their current devices, explore compatible products, evaluate upgrades, or assess resale opportunities.

D. Second-Life Buyers

Users interested in eligible previously owned products and the available evidence concerning their history and reported condition.

E. Second-Life Sellers

Eligible owners seeking to list previously purchased platform products for resale, subject to platform-defined validation and eligibility rules.

6.2 Project Stakeholders

- Falcon Team.
- DEPI program and project supervisors.
- Intended platform users.
- Product data providers and relevant third-party service providers, where applicable.
- Academic evaluators and project reviewers.

Specific external integrations and service providers remain subject to confirmation.

7. Core Features

The following six features constitute the currently confirmed core scope of TechWise.

7.1 AI Shopping Intelligence

Purpose: Help users express their needs in natural language and understand which products may satisfy them.

Intended Capabilities

- Interpret natural-language shopping requests.
- Extract structured requirements from user input.
- Identify explicit constraints and preferences.
- Assist users in discovering relevant products.
- Explain product differences in accessible language.
- Summarize relevant customer-review information when reliable review data is available.
- Explain recommendation factors using available product facts and decision results.
- Support follow-up questions and refinement of requirements.

Architectural Boundary

AI is responsible for language interpretation, requirement extraction, explanations, recommendation assistance, and review summarization.

AI is not the authoritative source of product specifications, prices, inventory, compatibility, purchase eligibility, or ownership status.

AI-generated requirements must be validated before they affect backend filtering, ranking, or business decisions.

Product recommendations must be grounded in available product records and applicable deterministic rules.

7.2 Verified Second-Life Marketplace

Purpose: Enable eligible products previously purchased through TechWise to enter a controlled resale workflow.

Intended Capabilities

- Identify products associated with platform purchase-origin records.
- Associate a physical device with a product asset record.
- Track ownership periods and ownership transfers.
- Validate the current owner's eligibility to list a product.
- Apply listing-state and dispute-related restrictions.
- Collect seller-reported condition information.
- Collect required evidence, including screenshots and a test video for electronics, according to the applicable listing requirements.
- Present available evidence and distinguish seller claims from platform-established facts.
- Support eligible second-life purchases and subsequent ownership transitions.

Eligibility Principles

Resale eligibility must be determined by backend validation rather than AI interpretation or seller assertions alone.

The validation process may include:

- Platform purchase or origin verification.
- Product asset identification.
- Current ownership verification.
- Active ownership status.
- Existing listing status.
- Applicable eligibility requirements.
- Blocking disputes or unresolved restrictions.

The exact purchase-reference format, evidence requirements, and eligibility implementation must be finalized during requirements and backend design.

Verification Transparency

The platform must distinguish between:

- **Seller Reported:** Information provided by the seller.
- **Evidence Provided:** Documents, screenshots, videos, or other submitted material.
- **Platform Confidence:** The level of confidence supported by the platform's actual verification process.

Submitting evidence does not, by itself, establish physical inspection or independent verification.

Physical platform verification must not be claimed unless a corresponding verification process is actually implemented.

7.3 Product Compatibility Graph

Purpose: Represent product and component relationships to support compatibility-aware discovery and decisions.

Intended Capabilities

- Represent relevant product and component relationships.
- Identify applicable compatibility requirements.
- Support compatibility-aware filtering.
- Help users evaluate whether a candidate product fits their existing setup.
- Explain compatibility results using structured facts and rules.

Proposed Data Concepts

Depending on the final product categories and technical requirements, compatibility relationships may include:

- Compatible with.
- Requires.
- Supports.
- Incompatible with.
- Alternative to.
- Upgrade path for.

These relationship types are proposed data-model concepts, not confirmed implemented entities or APIs.

Compatibility decisions must be based on structured records and deterministic rules wherever the requirements can be formally evaluated.

Unknown compatibility must not automatically be represented as confirmed compatibility.

7.4 Purchase Decision Engine

Purpose: Help users evaluate products against explicit requirements and compare alternatives systematically.

Intended Capabilities

- Accept structured user requirements.
- Apply mandatory constraints.
- Compare candidate products using relevant attributes.
- Evaluate trade-offs between alternatives.
- Consider budget, compatibility, specifications, and user priorities where applicable.
- Explain why products satisfy or fail specific requirements.
- Provide evidence-based decision summaries.

Decision Principles

The decision engine must distinguish mandatory constraints from preferences.

A product that violates a mandatory requirement must not be presented as fully compliant merely because it performs well in other dimensions.

The final ranking criteria, weighting system, tie-breaking rules, and treatment of missing product data must be documented before implementation.

AI may explain the result but must not silently override deterministic eligibility or filtering rules.

7.5 Product Truth Layer

Purpose: Improve the reliability, traceability, and clarity of product information presented to users.

Intended Capabilities

- Maintain structured product facts.
- Distinguish verified records from unverified claims.
- Associate facts with available evidence or data sources where supported.
- Identify missing or conflicting information.
- Present specifications consistently across product listings and comparisons.
- Support trustworthy explanations and decision-making.

Information Categories

The platform should distinguish among:

- Structured product data.
- Source-provided information.
- Seller-reported claims.
- Submitted supporting evidence.
- Platform-verified information, where verification actually exists.
- Unknown or unavailable information.

The exact confidence model, provenance schema, and verification rules remain to be finalized.

AI-generated content must not be treated as authoritative evidence of product facts.

7.6 Buy vs Upgrade vs Resell Engine

Purpose: Help users evaluate possible next steps for products they already own or are considering purchasing.

Intended Capabilities

- Evaluate whether purchasing a new product addresses the user's needs.
- Compare new-product options against the current device or setup.
- Identify relevant upgrade opportunities.
- Evaluate eligible resale opportunities.
- Compare available alternatives using transparent criteria.
- Explain the trade-offs of each option.

Decision Options

The engine may compare:

7. **Buy:** Purchase a new product.
8. **Upgrade:** Improve an existing product or setup.
9. **Resell:** Sell an eligible owned product.
10. **Keep:** Continue using the existing product when no change is justified.

The fourth option is a proposed extension to the decision workflow and requires explicit approval before becoming a confirmed feature.

Financial calculations, depreciation models, resale-value estimates, and recommendation thresholds must use documented rules and available data. They must not be fabricated by AI.

8. Cross-Cutting Capability: Advanced Smart Filtering

Advanced Smart Filtering is a shared capability across product discovery, product comparison, and decision support.

It is not a seventh core feature.

8.1 Objectives

- Reduce irrelevant search results.
- Enable users to refine product selections.
- Support category-specific attributes.
- Apply mandatory constraints consistently.
- Support compatibility-aware filtering when relevant data is available.

- Make filtering behavior predictable and transparent.

8.2 Potential Filter Dimensions

Subject to product category and data availability:

- Price range.
- Brand.
- Product category.
- Technical specifications.
- Product condition.
- Compatibility.
- Relevant user requirements.
- Availability, if inventory data is integrated.
- Evidence or verification status, where applicable.

8.3 Filtering Principles

- Filtering must be executed by deterministic backend logic.
- The frontend must accurately represent the active filters.
- Unsupported filters must not imply that corresponding data exists.
- Missing values must be handled explicitly.
- Filter options should reflect the applicable product category and available data.
- AI may translate natural-language requests into structured filter criteria, which must be validated before execution.

9. Product Lifecycle Model

TechWise is designed around a product lifecycle rather than a sequence of isolated shopping transactions.

Lifecycle:

Product → Product Asset → Purchase → Ownership → Usage → Review → Upgrade/Resell → Second-Life → New Ownership

9.1 Product

Represents a product model, catalog entry, or SKU.

Examples of attributes may include manufacturer, model, category, and technical specifications.

9.2 Product Asset

Represents a specific physical device.

A product asset is distinct from the general product model and may be associated with its own lifecycle history.

9.3 Purchase

Represents a purchase transaction or platform purchase record.

The purchase record provides the basis for establishing platform origin where applicable.

9.4 Ownership

Represents a specific user's ownership period for a product asset.

A physical device may have multiple ownership periods throughout its lifetime.

9.5 Usage and Review

Usage and review information may support lifecycle decisions, provided that its source, ownership, and privacy implications are handled appropriately.

9.6 Upgrade and Resale

Eligible owners may evaluate upgrade options or initiate a resale workflow subject to platform rules.

9.7 New Ownership

When an eligible resale is completed, the product asset remains the same physical device, while its ownership relationship changes.

Lifecycle Invariants

- Product age does not reset after resale.
- Ownership changes; the product asset remains the same.
- Original purchase history must remain associated with the asset where records permit.
- Previous ownership records must be retained as appropriate for lifecycle continuity.
- Previous owners' personal information must not be exposed to subsequent owners.
- A new owner must not be able to use an old owner's active ownership credentials as their own.
- Any new ownership reference must preserve the distinction between original purchase history and the current ownership period.

The exact reference-code mechanism and data-retention rules require implementation-level specification.

10. Functional Requirements

The following requirements describe intended platform behavior. They are not claims that the functionality has already been implemented.

ID	Requirement
FR-01	The platform shall provide product discovery and browsing capabilities.
FR-02	The platform shall support structured product information and product detail views.
FR-03	The platform shall allow users to compare relevant product attributes.
FR-04	The platform shall support advanced filtering using available product data.
FR-05	The platform shall support AI-assisted interpretation of shopping requirements.
FR-06	The platform shall validate AI-extracted criteria before applying backend rules.

ID	Requirement
FR-07	The platform shall support requirement-based purchase decisions.
FR-08	The platform shall evaluate product compatibility using structured information and deterministic rules.
FR-09	The platform shall distinguish verified information from unverified claims and unavailable data.
FR-10	The platform shall support purchase-related workflows within the implemented e-commerce scope.
FR-11	The platform shall associate eligible product assets with applicable platform purchase-origin records.
FR-12	The platform shall validate current ownership and resale eligibility before allowing eligible listings.
FR-13	The platform shall support the submission of required seller evidence for applicable resale listings.
FR-14	The platform shall preserve product asset lifecycle continuity across ownership transfers.
FR-15	The platform shall protect previous owners' personal information.
FR-16	The platform shall support evaluation of buying, upgrading, and reselling options where sufficient data is available.
FR-17	The platform shall provide a responsive user experience across supported desktop and mobile screen sizes.
FR-18	The platform shall apply appropriate authentication and authorization controls to protected operations, subject to the final security design.
FR-19	The platform shall handle invalid inputs, unavailable information, and failed operations predictably.
FR-20	The platform shall maintain appropriate records for implemented transactions, ownership changes, and eligibility decisions.

The final requirements specification must identify priorities, acceptance criteria, dependencies, and implementation status for each requirement.

11. Non-Functional Requirements

11.1 Usability

The platform should make complex product information understandable through consistent layouts, clear terminology, and explainable decisions.

11.2 Responsiveness

Interfaces should adapt to desktop and mobile screen sizes without losing essential content or functionality.

11.3 Performance

Product discovery, filtering, comparison, and other frequently used workflows should be evaluated against measurable response-time targets established during implementation planning.

No numerical performance guarantees are claimed in this proposal.

11.4 Security

The implementation should protect user accounts, ownership records, transaction-related information, and private seller evidence through appropriate authentication, authorization, validation, and access controls.

11.5 Privacy

Personal information must not be disclosed to unauthorized users. Lifecycle history should preserve necessary product information without exposing previous owners' personal data.

11.6 Data Integrity

Product facts, purchase records, ownership transitions, eligibility checks, and transaction-related changes must follow consistent backend rules.

11.7 Maintainability

The implementation should separate interface concerns, API handling, business logic, services, and data access to support testing and future changes.

11.8 Reliability

The platform should handle errors, incomplete records, invalid requests, and conflicting data without silently producing misleading results.

11.9 AI Reliability

AI-generated interpretations and explanations must be grounded in available data. Deterministic validation must remain authoritative for business rules and factual decisions.

11.10 Accessibility

The team should evaluate readable typography, adequate contrast, meaningful control labels, keyboard interaction where applicable, and understandable error messages.

12. UI/UX Design Direction

12.1 Confirmed Visual Direction

The currently established design direction combines:

- Glassmorphism.
- Organic Design.
- Dark Mode.
- A predominantly monochromatic palette based on white, black, and their shades.
- Large rounded and squircle-inspired shapes.
- Minimal, premium visual presentation.
- Responsive desktop and mobile layouts.

- Consistent typography, spacing, controls, and component behavior.

The approximately 50% radius preference refers to a visual direction toward highly rounded, pill-shaped, circular, or squircle-inspired elements where appropriate. It must not be interpreted as a universal CSS border-radius value for every component.

12.2 Design Principles

Clarity

Every screen should have a clear purpose and an understandable information hierarchy.

Consistency

Shared components should use consistent visual and interaction rules.

Functional Minimalism

Decorative content must not distract from the user's task.

Responsive Behavior

Desktop and mobile layouts must be designed intentionally rather than relying exclusively on automatic scaling.

Trust and Transparency

Product facts, seller claims, evidence, and decision explanations should be distinguishable.

Readability

Navigation labels and button text should remain on one line wherever specified by the established design rules. Essential labels must not be wrapped into two-line button or navigation labels.

Navigation

Floating, rounded navigation containers may use margins around the viewport and horizontally arranged controls according to the established visual direction.

12.3 Typography

Plus Jakarta Sans has been identified in the previous design-system direction. Its final usage, font weights, language support, and fallback fonts must be verified against the current design files.

12.4 Content Quality

The interface must not contain meaningless pseudo-technical filler, fake system telemetry, decorative coordinates, or irrelevant labels that do not serve a real user need.

12.5 Design Deliverables

The design phase should provide:

- A coherent design system.
- Shared UI components.
- Desktop and mobile screen designs.
- Navigation and interaction patterns.
- Core user flows.
- Screen inventory and coverage tracking.
- Design validation and handoff documentation.

12.6 Design Status

Earlier project conversations reported a substantial screen-design deliverable, including an earlier count of 89 desktop and mobile designs.

This count is historical and must not be treated as the final approved inventory until it is reconciled with the latest design report.

Final screen counts, completed screens, responsive variants, and outstanding design tasks must be confirmed against the latest available design source.

13. High-Level User Workflows

The following workflows describe intended behavior. Exact screens, APIs, and state transitions remain subject to the approved requirements and implementation.

13.1 AI-Assisted Shopping

11. The user expresses a shopping need.
12. The system interprets the request into structured requirements.
13. The backend validates the extracted requirements.
14. Deterministic filtering and decision logic evaluate available product records.
15. The system presents relevant candidates.
16. The user compares alternatives and reviews explanations.
17. The user proceeds to the applicable purchase workflow.

13.2 Product Comparison

18. The user discovers or searches for products.
19. The user selects products for comparison.
20. The system retrieves relevant structured attributes.
21. The system presents differences and missing information.
22. The user evaluates trade-offs.
23. The user proceeds with an informed decision.

13.3 Compatibility Evaluation

24. The user identifies an existing product or setup.
25. The system retrieves available product and compatibility information.
26. Deterministic rules evaluate known compatibility constraints.
27. The system presents compatible, incompatible, or unknown results as supported by the data.
28. The user reviews the explanation and available alternatives.

13.4 Eligible Second-Life Listing

29. The current owner initiates a resale workflow.
30. The backend identifies the associated product asset and platform-origin records.
31. The system validates current ownership and applicable eligibility conditions.
32. The seller provides required condition information and supporting evidence.
33. The system validates the listing requirements.
34. The listing becomes available only when the required conditions are satisfied.
35. An eligible transaction may trigger an ownership transition under the final transaction rules.

13.5 Buy vs Upgrade vs Resell

36. The user identifies a current need or an existing product.
37. The system retrieves available product, compatibility, ownership, and market information.
38. The decision engine evaluates applicable alternatives.
39. The system explains the benefits, limitations, and data uncertainty of each option.
40. The user decides whether to buy, upgrade, resell, or continue using the existing product, where these options are supported.

14. High-Level System Architecture

TechWise will follow a layered architecture that separates user interaction from authoritative business operations.

14.1 Logical Layers

Layer 1: User Interface

Responsible for presenting product information, accepting user input, managing interface state, and displaying results.

Layer 2: API Layer

Responsible for receiving requests, validating request structure, enforcing applicable access rules, and returning structured responses.

Layer 3: Business Logic

Responsible for enforcing business rules, validating workflows, evaluating eligibility, and coordinating deterministic operations.

Layer 4: Services

Responsible for reusable capabilities such as product comparison, compatibility evaluation, AI integration, and lifecycle operations where implemented.

Layer 5: Data Access

Responsible for controlled access to persisted records.

Layer 6: Database

Responsible for authoritative persisted product, purchase, ownership, listing, and other supported application records.

14.2 AI Processing Flow

User Intent → AI Interpretation → Structured Data → Validation → Backend Logic → Database → Result

AI output must be treated as untrusted input until validated.

14.3 Architectural Principles

- AI is not the source of truth.
- Product and ownership records are authoritative within their defined scope.
- Business rules must be enforced by backend logic.
- Authorization must be checked on protected operations.
- AI must not directly bypass validation or write arbitrary business state.
- Missing or conflicting data must be handled explicitly.

- The architecture must support testing of deterministic logic independently of AI output.

14.4 Technology Stack Status

The final implementation stack has not been confirmed in the information consolidated for this document.

The proposal therefore does not assert a specific frontend framework, backend framework, database engine, AI provider, payment provider, hosting platform, or authentication service.

The approved stack must be added after the team confirms its technology decisions.

15. Conceptual Data Model

The following are conceptual data entities for requirements and architecture discussions. They are not assertions that corresponding database tables, classes, or APIs already exist.

Concept	Purpose
User	Represents a platform account.
Product	Represents a catalog product model or SKU.
Product Specification	Stores structured attributes relevant to a product.
Product Asset	Represents a specific physical device.
Purchase	Represents an applicable purchase record.
Ownership Period	Represents a user's ownership of a specific product asset over a defined period.
Product Compatibility	Represents a compatibility relationship or rule.
Product Evidence	Represents available supporting evidence associated with relevant product information or claims.
Product Review	Represents a review and its available source information.
Resale Listing	Represents a product asset offered for resale.
Transaction	Represents an applicable purchase or resale transaction.
Decision Request	Represents structured criteria submitted for a product decision.
Decision Result	Represents a deterministic evaluation and its supporting explanations.

The final entity definitions, identifiers, relationships, constraints, and persistence strategy must be established during detailed data modeling.

15.1 Important Relationship Rules

- A product model and a physical product asset are different concepts.
- A physical product asset may have multiple ownership periods.
- Purchase-origin information must remain distinguishable from current ownership.
- Ownership transitions must preserve applicable lifecycle history.
- Personal information belonging to former owners must not be exposed through public lifecycle records.
- Resale eligibility must be derived from authoritative records and applicable business rules.

16. Project Scope

16.1 In Scope

The intended project scope includes:

- E-commerce-oriented product discovery.
- Structured product information.
- AI-assisted shopping interactions.
- Product comparison.
- Advanced smart filtering.
- Product compatibility evaluation.
- Purchase decision support.
- Product information transparency.
- Platform-origin second-life marketplace workflows.
- Ownership and lifecycle continuity.
- Buy, upgrade, and resale decision support.
- Responsive desktop and mobile UI.
- Backend validation and business rules.
- Relevant data modeling and persistence.
- Testing, documentation, and project evaluation.

16.2 Out of Scope Unless Explicitly Approved

The following are not confirmed project commitments:

- Guaranteed access to commercial product databases.
- Guaranteed live inventory or market-price integrations.
- Guaranteed real-money payment processing.
- Physical inspection of every second-life product.
- Automated repair or refurbishment services.
- Guaranteed third-party marketplace integrations.
- A specific AI provider or model.
- A specific deployment infrastructure.
- Production-scale operational guarantees.
- A fully automated depreciation or resale-price prediction model.
- Any external service that has not been selected and validated by the team.

These items may be evaluated as future enhancements or explicitly added after feasibility review.

17. Project Methodology and Work Plan

The project will progress through incremental planning, design, implementation, integration, and validation.

The schedule below reflects the dates previously shared for the DEPI project phases. It should be reconciled with the official program schedule before final submission.

Phase	Start	End	Primary Deliverables
Planning and Project Management	2 October 2026	16 October 2026	Proposal, planning artifacts, scope, roles, risks, KPIs

Phase	Start	End	Primary Deliverables
Literature Review and Requirements	9 October 2026	16 October 2026	Research findings, requirements, scope validation
Design	9 October 2026	6 November 2026	UI/UX designs, design system, architecture and data-model work as scheduled
Implementation	16 October 2026	30 November 2026	Implemented application features and integration
Final Testing, Reports, and Documentation	16 October 2026	4 December 2026	Testing evidence, final documentation, evaluation, submission

These dates overlap because the planned phases are intended to progress in parallel where appropriate.

17.1 Proposed Milestones

Milestone	Target Date	Completion Evidence
M1 — Proposal and Initial Planning	16 October 2026	Reviewed proposal and planning artifacts
M2 — Requirements Baseline	16 October 2026	Documented functional and non-functional requirements
M3 — Design Baseline	6 November 2026	Reviewed screen inventory, design system, and core user flows
M4 — Implementation Completion Target	30 November 2026	Integrated implementation and recorded test results
M5 — Final Testing and Submission	4 December 2026	Final reports, documentation, and evaluation evidence

The milestone dates are planning targets, not claims of completion or formal approval.

18. Team Organization and Collaboration

Falcon Team is responsible for the coordinated delivery of TechWise.

The team has previously discussed a six-person working structure. The final roster, individual assignments, and named ownership must be confirmed before submission.

18.1 Collaboration Principles

- Maintain one shared project repository.
- Use one working branch per team member alongside the main branch, according to the agreed branching plan.
- Avoid introducing a separate develop branch unless the team explicitly changes this decision.
- Coordinate integration through reviewed changes and shared interfaces.
- Maintain consistent requirements, API contracts, data definitions, and terminology.
- Ensure all members contribute to implementation, testing, documentation, and integration where appropriate.
- Avoid isolating individuals so completely by technical domain that they cannot collaborate across the product.

- Record meaningful decisions and update project documentation as the implementation evolves.

18.2 Proposed Responsibility Matrix

The following matrix is a proposed starting point, not an approved assignment of named individuals.

Work Area	Primary Responsibility	Shared Contribution
Project Coordination	Team leader	All team members
AI Shopping Intelligence	Designated AI owner	Backend, UI, testing, and documentation contributors
Product Catalog and E-Commerce	Assigned feature owners	Shared UI/API integration
Compatibility and Decision Logic	Assigned feature owners	Product data and testing contributors
Second-Life and Ownership Lifecycle	Assigned feature owners	Backend, UI, security, and testing contributors
UI/UX and Frontend Integration	Shared feature ownership	All members contribute to responsive behavior and validation
Data Modeling and Backend Integration	Shared technical ownership	All members follow agreed contracts
Testing and Documentation	Shared responsibility	Each member documents and tests their contributions

Final names and individual assignments must be supplied by the team. No unconfirmed names or allocations should be invented.

19. Project Management and Communication

The project should use a shared workflow to maintain visibility of tasks, blockers, dependencies, and completed work.

19.1 Recommended Practices

- Maintain a prioritized task backlog.
- Assign each task an owner and a clear completion condition.
- Track dependencies between design, APIs, database structures, and implementation.
- Review integration status regularly.
- Record decisions that affect the architecture or business rules.
- Keep documentation aligned with the actual implementation.
- Resolve conflicting assumptions before integrating dependent features.
- Use pull requests or equivalent review practices for meaningful changes.
- Record testing results and unresolved issues.

19.2 Repository Practices

Repository organization, naming conventions, branch protection, automated checks, and merge requirements must be finalized by the team.

The repository URL is established, but its contents, access permissions, and configuration must be verified during project setup.

20. Risk Management

The following risks are relevant to the proposed platform. Likelihood and impact ratings are preliminary assessments to be reviewed by the team.

Risk	Likelihood	Impact	Mitigation
AI-generated misinformation	Medium	High	Ground responses in product records; validate structured outputs; enforce backend rules.
Incomplete or inconsistent product data	High	High	Define required attributes, provenance, missing-data behavior, and validation rules.
Unclear compatibility information	Medium	High	Represent unknown states explicitly and use documented deterministic rules.
Second-life ownership fraud	Medium	High	Validate platform origin, current ownership, listing state, and eligibility server-side.
Misleading seller condition claims	Medium	High	Separate seller reports from evidence and verified facts.
Exposure of previous owners' information	Low to Medium	High	Enforce authorization and privacy controls; restrict access to historical personal data.
Scope expansion	High	Medium to High	Maintain the approved core scope and evaluate additions against schedule and capacity.
Integration conflicts	Medium	High	Agree on API contracts, data definitions, and shared interfaces before dependent implementation.
Delays in UI or backend delivery	Medium	High	Track dependencies, define milestones, and integrate incrementally.
Insufficient testing time	Medium	High	Define acceptance criteria early and test continuously.
Unavailable external services or data	Medium	Medium to High	Identify dependencies early and document fallback behavior.
Unclear ownership of team tasks	Medium	Medium	Confirm individual assignments and shared responsibilities.

Risk	Likelihood	Impact	Mitigation
Unsupported performance claims	Medium	Medium	Establish measurable targets and report only verified results.

The risk register must be updated as the project evolves.

21. Quality Assurance and Testing Strategy

Testing will be integrated throughout implementation rather than postponed until the final phase.

21.1 Unit Testing

Validate isolated business rules, calculations, input validation, filtering, ranking, compatibility rules, and eligibility decisions.

21.2 Integration Testing

Validate interactions between API endpoints, services, data access, and persistence.

21.3 AI Validation Testing

Evaluate natural-language requirement extraction, structured output validity, handling of ambiguous requests, and resistance to unsupported factual claims.

AI output must not bypass backend validation.

21.4 Lifecycle Testing

Verify that product assets retain their identity and lifecycle history across ownership transitions.

21.5 Authorization and Privacy Testing

Verify that users cannot access protected ownership records, submit unauthorized listings, or view another user's private information.

21.6 UI and Responsive Testing

Validate important workflows across supported desktop and mobile layouts, including navigation, form behavior, loading states, error states, and readability.

21.7 Acceptance Testing

Each implemented feature should be evaluated against documented acceptance criteria.

21.8 Evidence Requirements

Testing claims must be supported by actual test results, reproducible checks, or recorded demonstrations.

No test coverage percentage, performance result, security certification, or successful implementation milestone is claimed in this proposal without supporting evidence.

22. Key Performance Indicators

The following KPIs are proposed evaluation measures. Targets must be agreed upon before testing and must not be presented as achieved results without evidence.

KPI	Measurement Approach
Product Discovery Success	Percentage of test tasks in which users identify suitable candidate products.
Filtering Correctness	Percentage of evaluated filtering cases that match expected deterministic results.
Compatibility Accuracy	Percentage of tested compatibility cases correctly classified against the reference rules.
Decision Rule Correctness	Percentage of decision scenarios producing results consistent with documented constraints and criteria.
Product Information Traceability	Percentage of evaluated factual claims linked to an available source or documented evidence.
AI Requirement Extraction	Percentage of evaluated requests converted into valid structured criteria.
AI Unsupported-Claim Rate	Frequency of unsupported factual claims in the defined evaluation set.
Second-Life Eligibility Correctness	Percentage of test cases in which eligibility validation matches the expected outcome.
Lifecycle Continuity	Percentage of tested ownership transitions preserving required asset and purchase-history invariants.
Responsive Workflow Completion	Percentage of selected user workflows completed successfully on supported desktop and mobile layouts.
Critical Defect Resolution	Number of unresolved critical defects at the agreed release checkpoint.
User Satisfaction	Results from a documented user feedback questionnaire, if conducted.

Numerical targets, test-set sizes, evaluation procedures, and baseline values remain pending team approval.

23. Ethical, Privacy, and Trust Considerations

TechWise handles product information, purchase-origin data, ownership records, seller evidence, and potentially sensitive account information.

The platform should follow these principles:

- Collect only information necessary for supported functions.
- Restrict private records to authorized users.
- Avoid exposing previous owners' personal details.
- Clearly distinguish seller claims from platform-verified information.
- Avoid representing missing data as verified facts.
- Explain important decision criteria where feasible.

- Prevent AI-generated content from overriding authoritative business records.
- Define appropriate retention and deletion rules for personal information and submitted evidence.
- Evaluate applicable legal, privacy, and consumer-protection obligations before deployment.

The project must not claim compliance certifications or legal approval that have not been independently established.

24. Expected Deliverables

The project is expected to produce the following deliverables, subject to the approved scope and official submission requirements.

24.1 Planning and Analysis

- Project Proposal.
- Problem statement and objectives.
- Scope definition.
- Functional and non-functional requirements.
- Stakeholder and user analysis.
- Project schedule and milestones.
- Responsibility matrix.
- Risk register.
- KPI and evaluation plan.

24.2 Design

- UI/UX design system.
- Screen inventory.
- Desktop and mobile screen designs.
- Core user flows.
- Responsive design specifications.
- Architecture documentation.
- Conceptual data model.

24.3 Implementation

- Source code in the project repository.
- Implemented features within the approved scope.
- API and data-model documentation as applicable.
- Integration and configuration documentation.
- Setup instructions.

24.4 Verification

- Unit and integration test results.
- Feature acceptance results.
- Responsive testing evidence.
- AI evaluation results where applicable.
- Lifecycle and authorization testing evidence.
- Documented unresolved limitations.

24.5 Final Submission

- Final project report.
- Updated technical documentation.
- Demonstration materials.
- Final repository state.
- Presentation materials, if required by DEPI.
- Final evaluation results.

A deliverable is considered completed only when its completion is supported by the relevant artifact or evidence.

25. Current Project Status and Evidence Boundaries

This section separates established project decisions from implementation claims.

25.1 Confirmed Project Decisions

- The official project name is TechWise.
- The official description is Smart E-Commerce & Product Intelligence Platform.
- The team is Falcon Team.
- The platform vision follows the complete product lifecycle.
- The six core features are AI Shopping Intelligence, Verified Second-Life Marketplace, Product Compatibility Graph, Purchase Decision Engine, Product Truth Layer, and Buy vs Upgrade vs Resell Engine.
- Advanced Smart Filtering is a cross-cutting capability.
- AI is not the source of truth.
- Product records and deterministic backend logic govern factual and business decisions.
- Product, Product Asset, Purchase, and Ownership are distinct concepts.
- Product age must not reset after resale.
- Previous owners' personal information must not be exposed.
- The current design direction is Glassmorphism, Organic Design, Dark Mode, and a monochromatic primary palette.
- Desktop and mobile layouts are required design considerations.
- The repository URL is https://github.com/DEPI-5-Falcon-Project/TechWise_E-Commerce.

25.2 Previously Reported Design Progress

Earlier project conversations reported extensive UI/UX design work and a historical count of 89 desktop and mobile designs.

The latest design count, screen-by-screen completion status, and approval state remain to be verified against the latest design artifacts.

25.3 Information Not Yet Verified for This Document

The following information must not be fabricated:

- Final team member roster and named responsibility assignments.
- Final approved technology stack.
- Actual database schema.
- Implemented API endpoints.
- Confirmed AI provider and model.
- Confirmed payment and external integration providers.

- Final approved screen inventory.
- Current source-code contents and implementation progress.
- Actual test results and coverage.
- Final numerical KPI targets.
- Formal approval and supervisor sign-off.
- Confirmed deployment environment.

25.4 Repository Evidence

The repository URL is recorded as the official project reference.

Repository content, branch configuration, commits, source files, and implementation status must be inspected before making claims about completed code.

A repository link alone does not prove that a feature has been implemented.

26. Limitations and Assumptions

The proposal establishes the intended product direction and high-level requirements. It does not replace detailed technical specifications.

The following assumptions require confirmation during implementation planning:

- Product categories and required technical attributes.
- Product data sources and data acquisition methods.
- User authentication and authorization approach.
- Final API contracts.
- Database technology and detailed schema.
- AI provider, model, and evaluation approach.
- Product compatibility rule sources.
- Purchase and payment workflow scope.
- Resale transaction and ownership-transfer behavior.
- Evidence storage, access, and retention rules.
- Deployment and operational requirements.
- Final testing thresholds and measurable acceptance criteria.

Where an assumption cannot be validated, it must remain explicitly unresolved until the team makes a decision.

27. Conclusion

TechWise proposes a unified approach to technology e-commerce by combining product discovery, AI-assisted shopping, structured product intelligence, compatibility evaluation, purchase decision support, and lifecycle-aware second-life commerce.

Its central value is the connection between informed purchasing decisions and ongoing product ownership. Rather than ending the user journey at checkout, TechWise aims to preserve relevant product history and support future decisions involving usage, upgrades, resale, and ownership transfer.

The project will prioritize trustworthy product information, deterministic business rules, responsible AI integration, privacy, explainability, and responsive user experience.

The next steps are to finalize requirements, verify the current design inventory, confirm the technical stack and team responsibilities, establish the detailed architecture and data model, and begin implementation with measurable acceptance criteria.

This proposal is a consolidated planning document. Final approval, implementation status, and verification results must be recorded as the project progresses.

Appendix A — Glossary

Term	Definition
TechWise	Smart E-Commerce & Product Intelligence Platform.
AI Shopping Intelligence	AI-assisted interpretation and explanation of shopping requirements.
Product Truth Layer	The platform capability for presenting structured product facts with appropriate source and verification distinctions.
Product Compatibility Graph	Structured representation of product and component compatibility relationships.
Purchase Decision Engine	Deterministic evaluation of products against requirements and decision criteria.
Second-Life Marketplace	Marketplace for eligible previously owned products subject to platform rules.
Product	A catalog model or SKU.
Product Asset	A specific physical device with its own lifecycle.
Purchase	A transaction or purchase record.
Ownership Period	A specific user's period of ownership of a product asset.
Seller Reported	Information supplied by the seller that is not automatically verified.
Evidence Provided	Supporting material submitted to substantiate a claim.
Platform Confidence	Confidence supported by the platform's actual verification procedures and evidence.
Advanced Smart Filtering	A shared filtering capability across discovery, comparison, and decision workflows.
KPI	Key Performance Indicator used to evaluate project outcomes.